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#MAA155 Assignment INL1 - problem 1
#EL OTMANI Youssef (worked with CHARKI Omar)


def DIVINT(m,n): #input two integers m, n (respectively the divident and the divisor)
    print(m," divided by ", n )
    M = m
    N = n

    if(N == 0):
        return print("is impossible \n") # we can't divide by 0

    REM = M # the remainder
    QUOT = 0 # the quotient
    var = 0 # variable to identify each case

    if( M < 0 ):
        M = -M
        REM = -REM
        var = 1

    if( N < 0 ):
        N=-N
        var = var + 2

    while REM >= N:
        REM = REM - N
        QUOT = QUOT + 1


    if(var == 1) & (REM != 0):
        QUOT = - (QUOT + 1)
        REM = N - REM # the new remainder is still less than N

    if((REM == 0) & (var == 1)) | (var == 2):
        QUOT = - QUOT

    # there is nothing to do if var = 3 (both N and M are negative)

    return print("is ", QUOT, "and the remainder is ", REM, "\n" )



DIVINT(12,3)
DIVINT(-12,3)
DIVINT(12,-3)
DIVINT(-12,-3)

DIVINT(13,3)
DIVINT(-13,3)
DIVINT(13,-3)
DIVINT(-13,-3)

DIVINT(12,0)
DIVINT(-12,0)

DIVINT(0,3)
DIVINT(0,-3)

#BIRTHDAY
m=200501
n= 8
DIVINT(m,n)
DIVINT(m,-n)
DIVINT(-m,n)
DIVINT(-m,-n)
```

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12  divided by  3
is  4 and the remainder is  0

-12  divided by  3
is  -4 and the remainder is  0

12  divided by  -3
is  -4 and the remainder is  0

-12  divided by  -3
is  4 and the remainder is  0

13  divided by  3
is  4 and the remainder is  1

-13  divided by  3
is  -5 and the remainder is  2

13  divided by  -3
is  -4 and the remainder is  1

-13  divided by  -3
is  4 and the remainder is  1

12  divided by  0
is impossible

-12  divided by  0
is impossible

0  divided by  3
is  0 and the remainder is  0

0  divided by  -3
is  0 and the remainder is  0

200501  divided by  8
is  25062 and the remainder is  5

200501  divided by  -8
is  -25062 and the remainder is  5

-200501  divided by  8
is  -25063 and the remainder is  3

-200501  divided by  -8
is  25062 and the remainder is  5
```